EXPERIMENT 9

Aim: To understand Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers

Theory: Docker is a containerization platform that enables developers to package applications and their dependencies into lightweight, portable units called containers. Unlike traditional virtual machines, containers share the host operating system kernel, making them more efficient in terms of resource utilization and faster to start. Docker follows a client-server architecture, where the Docker client communicates with the Docker daemon to build, run, and manage containers.

The Docker architecture consists of key components such as Docker Client, Docker Daemon, Docker Images, and Docker Containers. The client sends commands (e.g., build, run, pull) to the daemon, which performs operations like creating and managing containers. Docker images are read-only templates used to create containers, while containers are running instances of these images. Docker also uses registries like Docker Hub to store and distribute images.

The container life cycle in Docker includes multiple stages: Create, Start, Run, Pause/Unpause, Stop, and Remove. A container is created from an image, started to execute the application, and can be paused or stopped when not in use. It can also be restarted or removed when no longer needed. By executing Docker commands, users can manage images and interact with containers efficiently, enabling consistent development, testing, and deployment across different environments.

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Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kashishparyani>docker --help
Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  bake        Build from a file
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login       Authenticate to a registry
  logout      Log out from a registry
  search      Search Docker Hub for images
  version     Show the Docker version information
  info        Display system-wide information

Management Commands:
  agent*      Docker AI Agent Runner
  ai*         Docker AI Agent - Ask Gordon
  builder     Manage builds
  buildx*     Docker Buildx
  compose*    Docker Compose
  container   Manage containers
  context     Manage contexts
  debug*      Get a shell into any image or container
  desktop*    Docker Desktop commands
  dhi*        CLI for managing Docker Hardened Images
  extension*  Manages Docker extensions
  image       Manage images
  init*       Creates Docker-related starter files for your project
  manifest    Manage Docker image manifests and manifest lists
  mcp*        Docker MCP Plugin
  model*      Docker Model Runner
  network     Manage networks
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  model*      Docker Model Runner
  network     Manage networks
  offload*    Docker Offload
  plugin      Manage plugins
  sandbox*    Docker Sandbox
  sbom*       View the packaged-based Software Bill Of Materials (SBOM) for an image
  scout*      Docker Scout
  system      Manage Docker
  volume      Manage volumes

Swarm Commands:
  swarm       Manage Swarm

Commands:
  attach      Attach local standard input, output, and error streams to a running container
  commit      Create a new image from a container's changes
  cp          Copy files/folders between a container and the local filesystem
  create      Create a new container
  diff        Inspect changes to files or directories on a container's filesystem
  events      Get real time events from the server
  export      Export a container's filesystem as a tar archive
  history     Show the history of an image
  import      Import the contents from a tarball to create a filesystem image
  inspect     Return low-level information on Docker objects
  kill        Kill one or more running containers
  load        Load an image from a tar archive or STDIN
  logs        Fetch the logs of a container
  pause       Pause all processes within one or more containers
  port        List port mappings or a specific mapping for the container
  rename      Rename a container
  restart     Restart one or more containers
  rm          Remove one or more containers
  rmi         Remove one or more images
  save        Save one or more images to a tar archive (streamed to STDOUT by default)
  start       Start one or more stopped containers
  stats       Display a live stream of container(s) resource usage statistics
  stop        Stop one or more running containers
  tag         Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE
  top         Display the running processes of a container
  unpause     Unpause all processes within one or more containers
  update      Update configuration of one or more containers
```

```
C:\Users\kashishparyani>docker --version
Docker version 29.4.0, build 9d7ad9f

C:\Users\kashishparyani>docker search ubuntu
NAME                              DESCRIPTION                                     STARS     OFFICIAL
ubuntu                            Ubuntu is a Debian-based Linux operating sys…   17813     [OK]
ubuntu/squid                      Squid is a caching proxy for the Web. Long-t…   125
ubuntu/nginx                      Nginx, a high-performance reverse proxy & we…   140
ubuntu/bind9                      BIND 9 is a very flexible, full-featured DNS…   117
ubuntu/kafka                      Apache Kafka, a distributed event streaming …   60
ubuntu/apache2                    Apache, a secure & extensible open-source HT…   102
ubuntu/zookeeper                  ZooKeeper maintains configuration informatio…   14
ubuntu/prometheus                 Prometheus is a systems and service monitori…   79
ubuntu/jre                        Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet              Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   73
ubuntu/postgres                   PostgreSQL is an open source object-relation…   43
ubuntu/python                     A chiselled Ubuntu rock with the Python runt…   32
ubuntu/redis                      Redis, an open source key-value store. Long-…   25
ubuntu/dotnet-runtime             Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps                Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                    Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                  Memcached, in-memory keyvalue store for smal…   6
ubuntu/jdk                        Ubuntu rock for OpenJDK                         0
ubuntu/loki                       Grafana Loki, a log aggregation system like …   2
ubuntu/mimir                      Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/rabbitmq                   Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/opentelemetry-collector    This is a rock for the OpenTelemetry Collect…   0
ubuntu/git-sync                   Clone a git repo and keep it in sync with th…   2
ubuntu/grafana-agent              Ubuntu ROCK for Grafana Agent, an open-sourc…   0

C:\Users\kashishparyani>docker search --filter=stars=1000 ubuntu
NAME      DESCRIPTION                                     STARS     OFFICIAL
ubuntu    Ubuntu is a Debian-based Linux operating sys…   17813     [OK]
```

```
C:\Users\kashishparyani>docker search mysql
NAME                    DESCRIPTION                                     STARS     OFFICIAL
mysql                   MySQL is a widely used, open-source relation…   16115     [OK]
bitnami/mysql           Bitnami Secure Image for mysql                  147
circleci/mysql          MySQL is a widely used, open-source relation…   32
bitnamicharts/mysql     Bitnami Helm chart for MySQL                    0
cimg/mysql                                                              3
ubuntu/mysql            MySQL open source fast, stable, multi-thread…   73
google/mysql            MySQL server for Google Compute Engine          26
elestio/mysql           Mysql, verified and packaged by Elestio         3
linuxserver/mysql       A Mysql container, brought to you by LinuxSe…   48
alpine/mysql            mysql client                                    7
docksal/mysql           MySQL service images for Docksal - https://d…   0
eclipse/mysql           Mysql 5.7, curl, rsync                          3
ilios/mysql             Mysql configured for running Ilios              1
datajoint/mysql         MySQL image pre-configured to work smoothly …   2
cleanstart/mysql        Secure by Design, Built for Speed, Hardened …   0
ddev/mysql              ARM64 base images for ddev-dbserver-mysql-8.…   2
mysql/mysql-server      Optimized MySQL Server Docker images. Create…   1034
mirantis/mysql                                                          0
corpusops/mysql         https://github.com/corpusops/docker-images/     0
cbioportal/mysql        This repository hosts MySQL database images …   1
vulhub/mysql                                                            1
mysql/mysql-router      MySQL Router provides transparent routing be…   29
mysql/mysql-cluster     Experimental MySQL Cluster Docker images. Cr…   100
vitess/mysql            Lightweight image to run MySQL with Vitess      1
mysql/mysql-operator    MySQL Operator for Kubernetes                   1
```

```
C:\Users\kashishparyani>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
b40150c1c271: Pull complete
92842f25412d: Download complete
Digest: sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41c7b
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu

C:\Users\kashishparyani>docker images
                                                                                    i Info →   U  In Use
IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
ubuntu:latest   c4a8d5503dfb        119MB         31.7MB
```

```
C:\Users\kashishparyani>docker run -it ubuntu
root@dd5dd302fecc:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@dd5dd302fecc:/# cd home
root@dd5dd302fecc:/home# ls
ubuntu
root@dd5dd302fecc:/home# cd ubuntu
root@dd5dd302fecc:/home/ubuntu# ls
root@dd5dd302fecc:/home/ubuntu# exit
exit
```

```
C:\Users\kashishparyani>docker run -d httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
4e5144803049: Pull complete
4f4fb700ef54: Pull complete
5435b2dcdf5c: Pull complete
a838a4be55ce: Pull complete
8261ede62f8c: Pull complete
61e1798e23ba: Pull complete
5148763a788d: Download complete
747d2a9231eb: Download complete
Digest: sha256:bdba5c86022f2d6ad0783168b07fe80541fe45e97b434b13ba8c959a5050fb6b
Status: Downloaded newer image for httpd:latest
e64661c705ec59864a403a273db7ed3280eb9d8f02572967aa40429279d86262

C:\Users\kashishparyani>docker run -d mysql
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
8faff6d2a6d2: Pull complete
c28152004afe: Pull complete
e639cd467a97: Pull complete
84e79c47b301: Pull complete
0c97bec1a348: Pull complete
e885080b838e: Pull complete
6497d062da4f: Pull complete
bb5107df7baa: Pull complete
664c1f1ffcc8: Pull complete
980d6178a910: Pull complete
6b65aba3f1f6: Download complete
1b038cfc7980: Download complete
Digest: sha256:c5df04bee1a42b74a5841c6409e669cf62126cd0416f00c1cea8ab933b9361b9
Status: Downloaded newer image for mysql:latest
8d226bb2db76401e13643ec4b93977d1cd7705d4a3f41345ecbfed5fdf7afeda

C:\Users\kashishparyani>docker run -d ubuntu
31c12af08cec14ccfeaf2097b7bbec315cb0d36506861db5dc0d117e351c2c8b
```

```
C:\Users\kashishparyani>docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED          STATUS          PORTS     NAMES
e64661c705ec   httpd     "httpd-foreground"   58 seconds ago   Up 57 seconds   80/tcp    recursing_allen

C:\Users\kashishparyani>
```

Conclusion: Successfully implemented Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers.